Signals & Systems

Communication-system project report

2025-Bylaw

Section 4

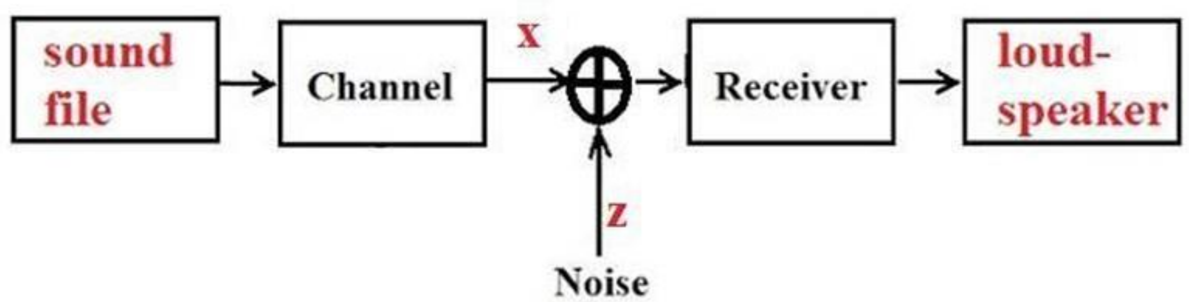
معاذ هشام عمرو 24010736

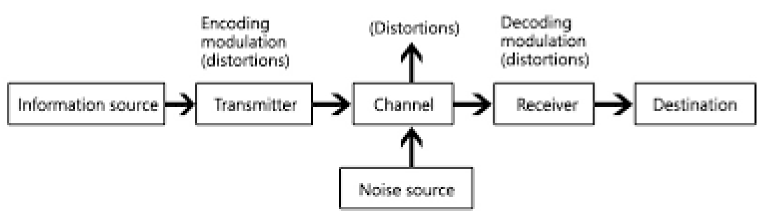
يوسف محمد حسن 24010866

محمد محمد عبدالمجيد محمد 24010648

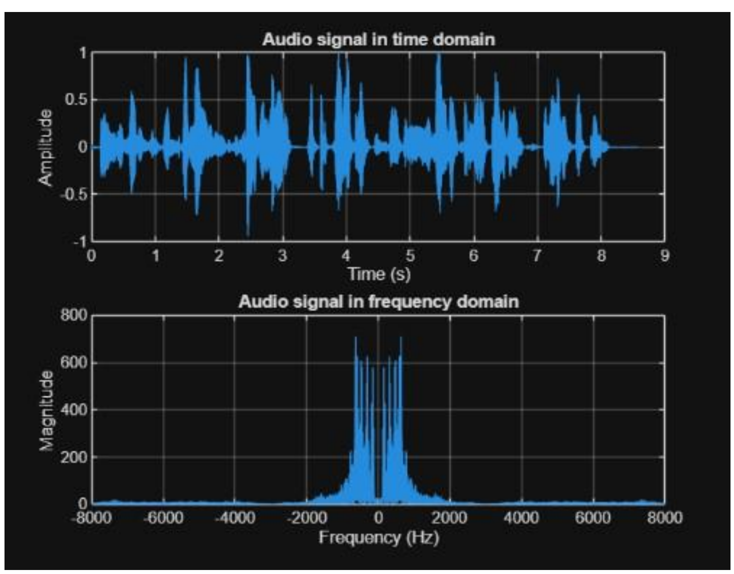
عمرو محمد السيد 2401163

In this project, a simple communication system was implemented using MATLAB.
An audio signal was transmitted through different communication channels, corrupted by Gaussian noise, then recovered using an ideal low-pass filter

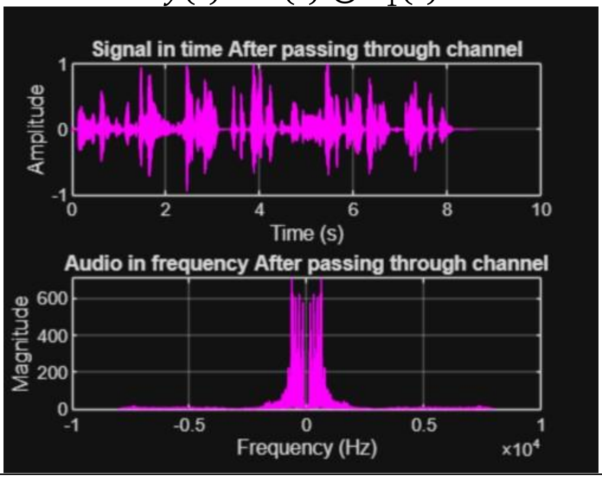
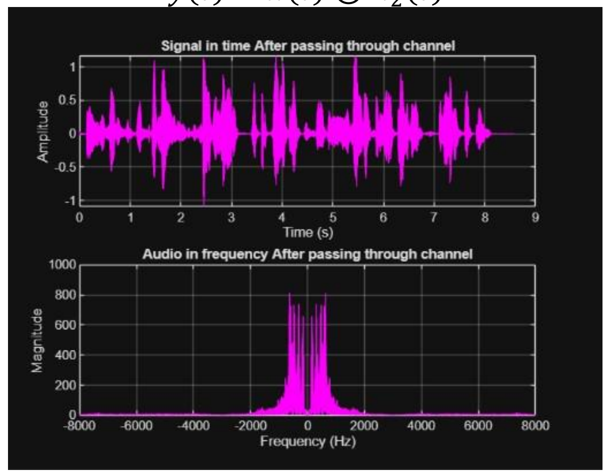
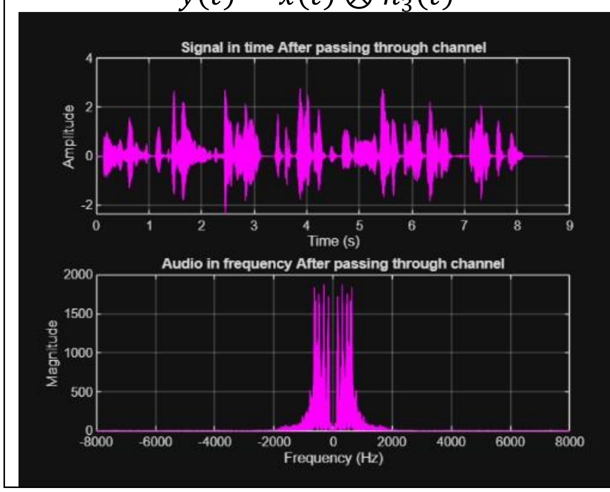
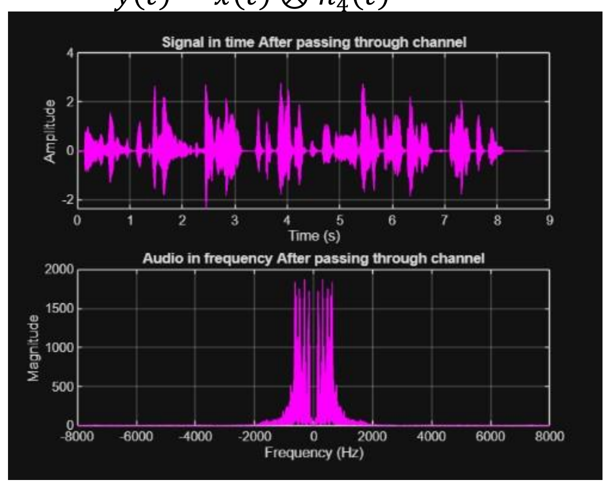




1-Transmitter



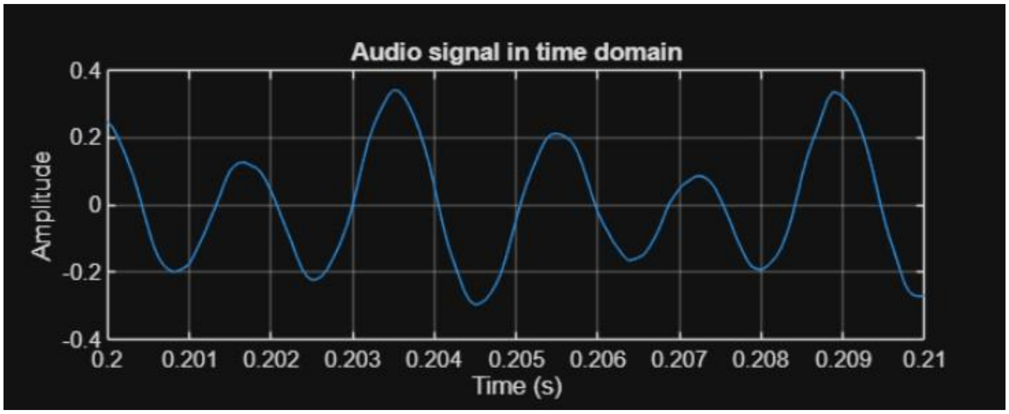
2-Channel

$h_1(t) = \delta(t)$ $y(t) = x(t) \otimes h_1(t)$  	$h_2(t) = e^{-10000\pi t}$ $y(t) = x(t) \otimes h_2(t)$ 
$h_3(t) = e^{-2000\pi t}$ $y(t) = x(t) \otimes h_3(t)$  	$h_4(t) = 2\delta(t) + 0.5\delta(t-1)$ $y(t) = x(t) \otimes h_4(t)$ 

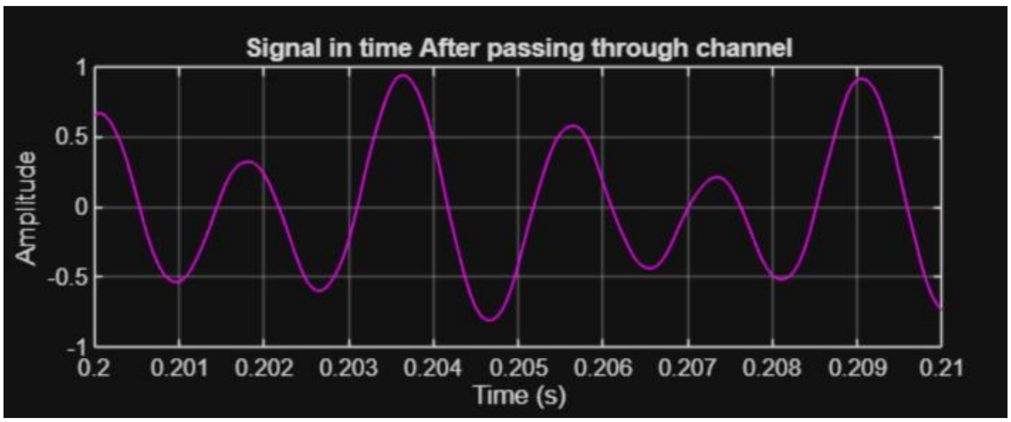
Observations:

- $h_1(t)$ doesn't change the signal shape as conv with delta gives the same function
- $h_2(t)$ causes slight smoothing and High frequencies attenuated slightly
- $h_3(t)$ causes strong smoothing and High frequencies attenuated strongly and causes lower sound quality

- Effect of $h_1(t)$ (sharp edges on time domain)



- Effect of $h_2(t)$ | $h_3(t)$ (smooth edges on time domain)

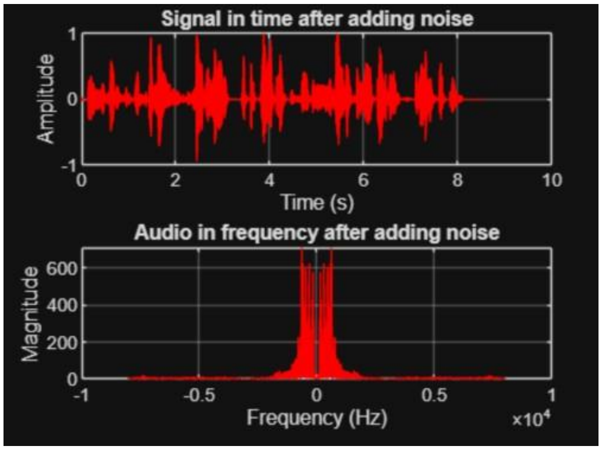


3-Noise

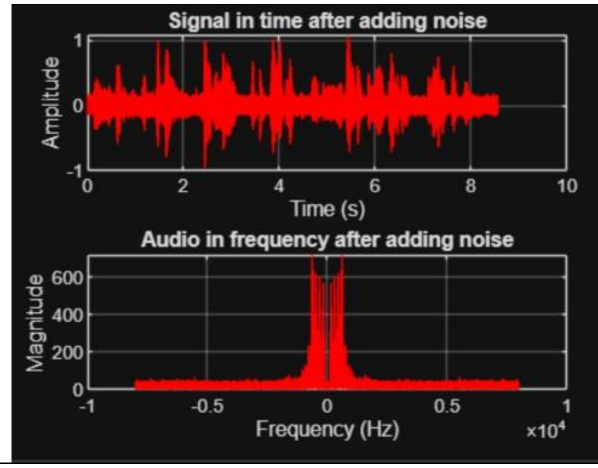
$$Z(t) = \sigma \cdot randn\big(1, length(x)\big)$$

- Sigma controls noise power , larger sigma --> stronger noise

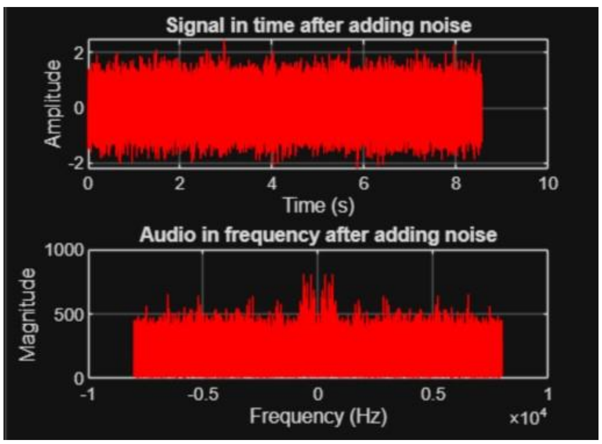
$\sigma = 0.001$



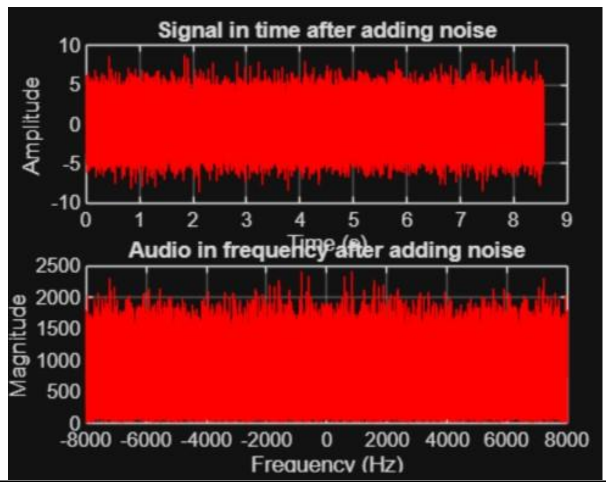
$\sigma = 0.05$



$\sigma = 0.5$



$\sigma = 2$



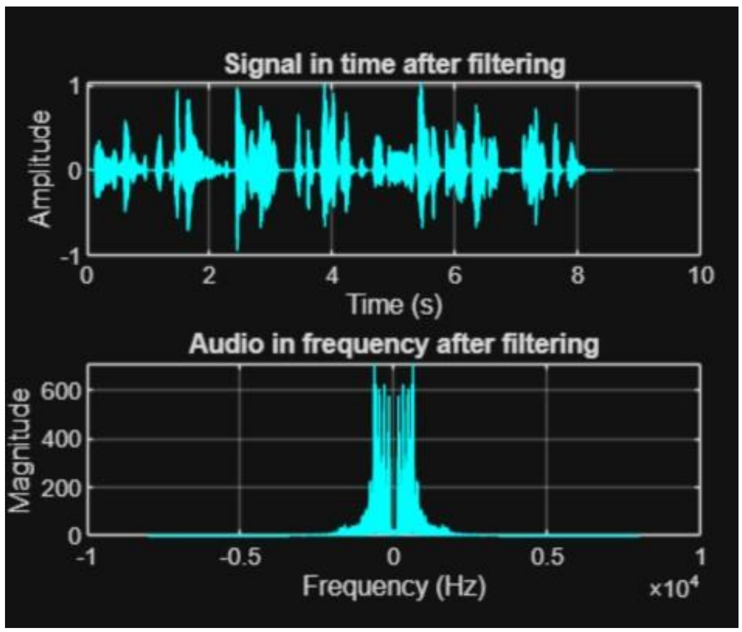
After adding Gaussian noise, the signal became distorted and extra frequency components appeared in the spectrum.

4-LPF

$$H(f) = \begin{cases} 1, & |f| \le f_c \\ 0, & |f| > f_c \end{cases}$$

- filter removes high-frequency noise
- improves received signal

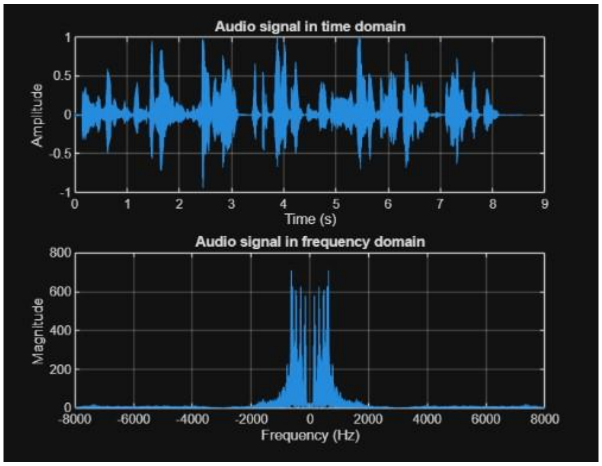
Plotting signal after passing through LPF



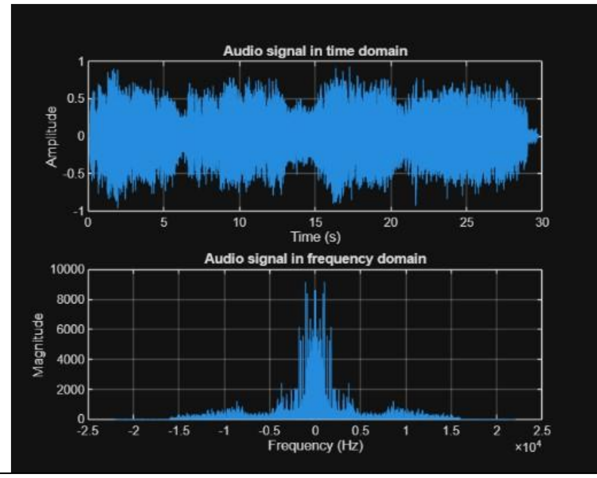
The low-pass filter reduced the noise significantly.
However, some high-frequency components of the music signal were also removed, causing a reduction in sound quality.

Comparison between Sound file vs Music file

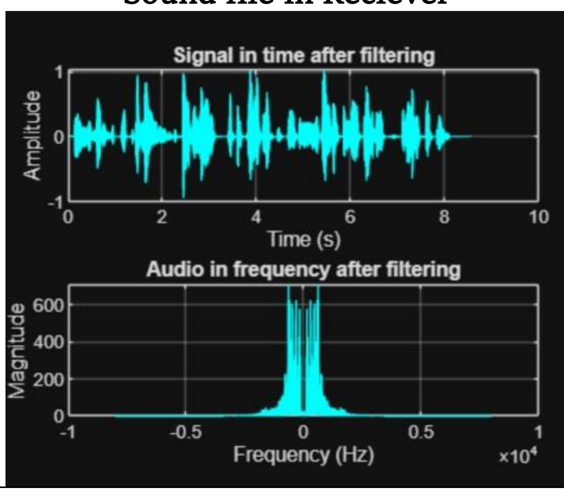
Sound file in Transmitter



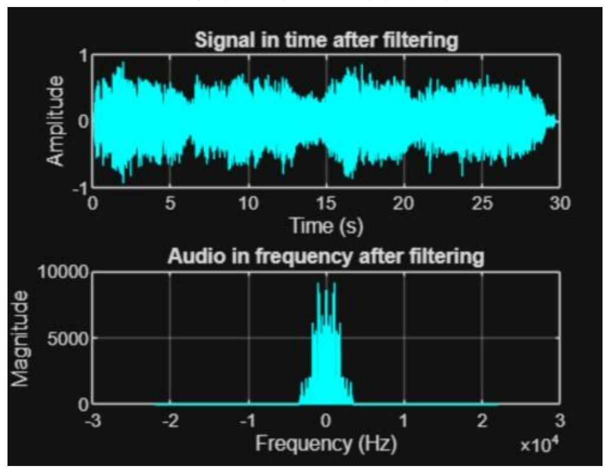
Music file in Transmitter



Sound file in Reciever



Music file in Reciever



The system performs better with speech signals than music signals because speech occupies lower bandwidth.